

# 1 Material consumptions MES --> ERP

## Overview

The system transfers the material withdrawals recorded in the MES including the return transfers to the ERP system.

In the ERP system the withdrawal postings are deducted from the warehouse and added to the individual production order. If withdrawals for an ERP batch cannot be executed in the production warehouse specified by the MES (missing inventory), the posting will be performed for that warehouse where the ERP batch is found. Only if this also fails will the system issue an error message. The ERP system should also use this approach (searching for an ERP batch) for return transfers.

The MES provides this data to the ERP system at regular intervals. The IDoc is of the type **ZWAU02**. This leads to the following IDoc specification:

|                                      |                                                    |
|--------------------------------------|----------------------------------------------------|
| Message type                         | <b>ZWAU</b>                                        |
| File name (for file transfers)       | <b>Z2WAU000X000</b>                                |
| File extension (for file transfer)   | Depending on the configuration (by default ".dat") |
| IDOC type (with tRFC communication): | <b>ZWAU02</b>                                      |
| Segments:                            | <b>Z2WAU000X000</b> (goods issues)                 |

Respect the following conventions if SAP is connected:



Create SAP segment names according to the pattern Z1<segment name> in order to generate the above-mentioned segment names in SAP. Versioning in SAP outbound processing then creates the segment names in the form Z2<Segment name><Version>.

Example: the created segment name **Z1WAU000X** is converted to **Z2WAU000X000**.

This documentation uses the below-mentioned column headings with the meanings described here:

| Column    | Description                                                                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Field     | Name of the field                                                                                                                                                                                         |
| V (usage) | <div>S Key field clearly identifying the data record. (Further key fields might be required). The field must be populated.</div> <div>M Mandatory field which must be populated with a valid value.</div> |

| Column           | Description                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------|
|                  | K Field may stay empty (optional field).                                                                          |
| T(type)          | Data type according to description.                                                                               |
| L(ength)         | Field length<br>For fields of data type DEC: Total number of digits without decimal separator and algebraic sign. |
| D(ecimal places) | For fields of data type DEC: Number of decimal places; otherwise: not relevant                                    |
| Description      | Field description and/or comments on the field.                                                                   |

## Movement types

In general, the following movement types are supported:

| Movement type | Description/ usage                               |
|---------------|--------------------------------------------------|
| 261           | Goods issue for production order                 |
| 262           | Cancellation of goods issue for production order |

## Data structure (Z2WAU000X000)

| Field      | T    | L  | D | Description                                                                                                                                                                                                                                                                                     |
|------------|------|----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WERK       | CHAR | 4  |   | Company/ plant/ site; stored in MES (fixed)                                                                                                                                                                                                                                                     |
| BEWART     | CHAR | 3  |   | Movement type (see table above)                                                                                                                                                                                                                                                                 |
| MATPUF     | CHAR | 12 |   | referring to batch: Material buffer according to batch inventory in MES<br>anonymous batch → input buffer of the machine number (MNR)                                                                                                                                                           |
| LGORT      | CHAR | 20 |   | ERP storage location<br>Storage location stored to the material buffer.                                                                                                                                                                                                                         |
| ATK        | CHAR | 40 |   | Material number                                                                                                                                                                                                                                                                                 |
| MENGE      | QUAN | 13 | 3 | Batch quantity/ consumption<br>The consumed quantity is always provided with a positive sign.                                                                                                                                                                                                   |
| MENGE_EINH | CHAR | 3  |   | Unit for batch quantity/consumption                                                                                                                                                                                                                                                             |
| ANR        | CHAR | 40 |   | Combined HYDRA production order number for which material has been withdrawn.<br>The exact length that is uploaded/confirmed depends on how the lengths are configured for the order or operation in the HYDRA basic parameter settings.<br>Used for ERP inbound processing if SAP is not used. |
| SAP_AUNR   | CHAR | 12 |   | SAP order number<br>Used for ERP inbound processing if SAP is in use.                                                                                                                                                                                                                           |
| SAP_AFOLG  | CHAR | 6  |   | SAP sequence number<br>Used for ERP inbound processing if SAP is in use.                                                                                                                                                                                                                        |
| SAP_VORNR  | CHAR | 4  |   | SAP operation number<br>Used for ERP inbound processing if SAP is in use.                                                                                                                                                                                                                       |
| SAP_UVGNR  | CHAR | 4  |   | SAP sub-operation number<br>Used for ERP inbound processing if SAP is in use.                                                                                                                                                                                                                   |
| KZEAUS     | CHAR | 1  |   | J - indicates whether this is the last withdrawal for this component. The system sets this ID for automatically logged off input batches when logging off operations. The ID indicates that no further consumption occurs for an operation-related component.                                   |
| HY_LOSNR   | CHAR | 20 |   | MES batch number (corresponding to this ERP batch).                                                                                                                                                                                                                                             |

| Field              | T    | L  | D | Description                                                                                                                  |
|--------------------|------|----|---|------------------------------------------------------------------------------------------------------------------------------|
| HY_DLLNR           | CHAR | 20 |   | MES throughput batch number (corresponding to this ERP batch)                                                                |
| ARBPL              | CHAR | 8  |   | Consuming MES machine                                                                                                        |
| GRND               | NUM  | 4  |   | Reserved (e.g. blocking reason when batch is logged off)                                                                     |
| LHW                | CHAR | 20 |   | Information on batch if there is a reference to the input batch                                                              |
| PERSNO             | NUM  | 8  |   | Personnel number of the person logged in to the terminal                                                                     |
| LST01              | QUAN | 13 | 3 | Consumption activity 1                                                                                                       |
| LST01_EINH         | CHAR | 3  |   | Unit of activity 1                                                                                                           |
| LST02              | QUAN | 13 | 3 | Consumption activity 2                                                                                                       |
| LST02_EINH         | CHAR | 3  |   | Unit of activity 2                                                                                                           |
| LST03              | QUAN | 13 | 3 | Consumption activity 3                                                                                                       |
| LST03_EINH         | CHAR | 3  |   | Unit of activity 3                                                                                                           |
| LST04              | QUAN | 13 | 3 | Consumption activity 4                                                                                                       |
| LST04_EINH         | CHAR | 3  |   | Consumption unit of activity 4                                                                                               |
| LST05              | QUAN | 13 | 3 | Consumption activity 5                                                                                                       |
| LST05_EINH         | CHAR | 3  |   | Unit of activity 5                                                                                                           |
| LST06              | QUAN | 13 | 3 | Consumption activity 6                                                                                                       |
| LST06_EINH         | CHAR | 3  |   | Unit of activity 6                                                                                                           |
| LOSSTATUS          | CHAR | 1  |   | Batch status when input batch is logged off                                                                                  |
| LOSKLASSE          | CHAR | 1  |   | Batch class when input batch is logged off                                                                                   |
| CHARGENNUMMER      | CHAR | 10 |   | ERP batch number                                                                                                             |
| CHARGENNUMMER_LONG | CHAR | 20 |   | ERP batch number (long)<br>Available from MPL 8.2 on - please also see the following information on field CHARGENNUMMER_LONG |
| MSL_VFDATE         | DATE | 8  |   | MSL expiry date<br>This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.               |
| MSL_VFTIME         | TIME | 6  |   | MSL expiry time<br>This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.               |
| MSL_PERIOD         | NUMC | 8  |   | MSL term<br>This field is only available if the database patch "dbp_mpl_mslmonitoring.hsc" is executed.                      |

#### Information on the fields CHARGENNUMMER / CHARGENNUMMER\_LONG

The field CHARGENNUMMER\_LONG is only available as of MPL 8.2.

If MPL 8.2 is used, the fields CHARGENNUMMER and CHARGENNUMMER\_LONG are populated as follows:

CHARGENNUMMER includes the ERP batch number with the characters 1-10.

CHARGENNUMMER\_LONG includes the ERP batch number with the characters 1-20.

You should use the value of the CHARGENNUMMER\_LONG field for new installations.